AUDIOSHIELD FORENSICS

Audio Forensic Analysis Report

Report ID	ASF-20260923-58FB
Generated	2026-09-23 22:56:14 India Standard Time
File Analyzed	b8387fa1-1ecb-4a57-baec-891c1051192d.m4a
Analyst	AudioShield Forensics System
System Version	AudioShield Forensics v1.0.0
Report Version	1.0.0
Classification	CONFIDENTIAL — For authorized use only

CHAIN OF CUSTODY — Automated generation, no manual modification
Statement: This report was generated automatically by the AudioShield Forensics System. No manual modification has been made to the analysis results. All evidence sources are independently computed and logged.
Integrity: Report content is derived solely from the pipeline outputs produced at the timestamp shown above. Any post-generation alteration would invalidate this document.

Source	Description
AI Detection	CNN (mel-spectrogram) · SVM (MFCC) · AASIST (raw waveform, graph attention)
Metadata	FFprobe (container / codec) + EXIFTool (embedded tags)
Signal	Spectral features (MFCCs, centroid, bandwidth) + temporal (RMS, silence ratio)
Fusion	Rule-based combiner — AI risk drives overall level; metadata and signal corroborate

This report presents separate, independently-sourced evidence streams and conservative forensic indicators derived from three AI models, metadata extraction, and signal analysis. It is not a substitute for expert examination of the original evidence and does not establish legal authenticity on its own.

Models trained on ASVspoof 2019 LA (English speech). Results are probabilistic indicators — not definitive judgements.

1. Executive Summary

Overall risk assessment: **medium**

Overall assessment: **inconclusive** | Confidence: **0.9234225157240048**

Key findings

- AI consensus: spoof (risk: high).
- Metadata risk: low; 2 finding(s).
- Signal risk: low; 0 anomaly/anomalies.

Official Eval AI comparison: CNN 69.83%, SVM 89.40%, AASIST 86.45%.

AASIST is the primary forensic model, achieving 99.79% precision and 7.20% EER on the official eval set.

Recommendation: Evidence is mixed or incomplete; requires further investigation.

2. File Information

Property	Value
File name	b8387fa1-1ecb-4a57-baec-891c1051192d.m4a
File size (bytes)	31875
MD5 hash	Not available
Format	mov,mp4,m4a,3gp,3g2,mj2
Duration (seconds)	6.528063
Sample rate (Hz)	16000
Channels	1
Codec	aac
Bitrate (bits/sec)	39062
Creation time	2026-09-23T17:01:28.000000Z
Modification time	2026:09:23 17:01:28

3. AI Detection Results

Official Eval-set comparison

Rank	Model	Accuracy	Precision	EER
1st	AASIST	86.45%	99.79%	7.20%
2nd	SVM	89.40%	95.44%	16.98%
3rd	CNN	69.83%	99.99%	7.16%

Ranking prioritizes forensic suitability: **precision first** (50%), then **EER** (35%), then accuracy (15%). Precision is the primary criterion because a false positive — flagging a genuine speaker as synthetic — carries the highest forensic cost. CNN ranks 3rd despite the highest precision (99.99%) because its recall of only 66.4% means it misses one-third of all spoof samples.

Best model: **AASIST** (best forensic suitability: highest precision 99.79%, low EER 7.20%, strong accuracy 86.45%). AASIST is recommended as the primary model due to its highest precision among well-performing models and the lowest false positive rate (1.54%).

4. Metadata Forensics

Extracted metadata

Field	Value
File Name	b8387fa1-1ecb-4a57-baec-891c1051192d.m4a
File Size	31875
Format	mov,mp4,m4a,3gp,3g2,mj2
Duration	6.528063
Codec	aac
Sample Rate	16000
Channels	1
Bitrate	39062
Encoder	Not available
Creation Time	2026-09-23T17:01:28.000000Z
Modification Time	2026:09:23 17:01:28
Device Info	Not available
Software	Not available
Comments	Not available

Risk score: **low**

Findings

Severity	Field	Description
low	device	Device information is missing; provenance cannot be verified from metadata alone.
low	encoder	Encoder or software information is missing.

Metadata indicators are not conclusive on their own and require further investigation.

5. Signal Forensics

Spectral features

Feature	Mean	Std	Min	Max
Spectral Centroid	1772.1890855057638	1267.8856824414622	0.0	5074.327179941491

Feature	Mean	Std	Min	Max
Spectral Bandwidth	1428.311658840135 6	607.3785144575628	0.0	2851.386462303061
Spectral Rolloff	3277.567675159235 5	2137.212835707652	0.0	7187.5
Spectral Flatness	0.18149975624455 808	0.34644052489505 295	9.069852239917964e-06	1.0000007152557373
Spectral Contrast	22.559685390920 94	5.535563888421251	10.737564707492076	32.52856533485 79

Temporal features

Feature	Mean	Std	Min	Max
Zero Crossing Rate	0.1515848925159 2358	0.1398852806091 2456	0.0	0.58056640625
Rms Energy	0.0242786831294 81643	0.0311844417085 3396	0.0	0.1166383624076 8433
Pitch	112.6924560028 3439	8.412881053217 031	91.43616857570 666	155.5634918610 4043

Risk score: **low**

Anomalies

No signal-level anomalies were reported.

6. Evidence Fusion

Evidence source	Risk / result
AI	spoof (risk: high)
Metadata	low
Signal	low
Overall	medium

Assessment: **inconclusive** | Confidence: **0.9234225157240048**

Reasoning: Source risks are AI=high, metadata=low, signal=low; applied documented rule-based logic.

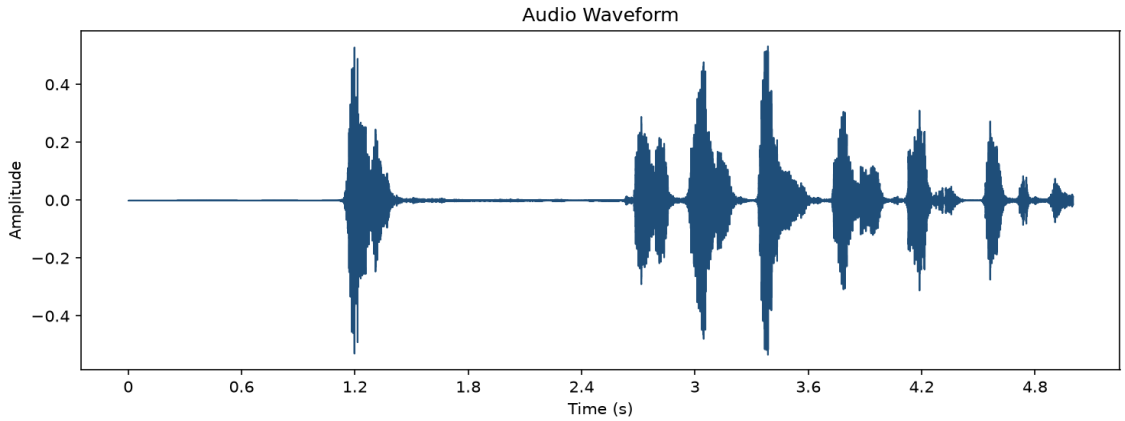
Recommendation: Evidence is mixed or incomplete; requires further investigation.

AI model breakdown

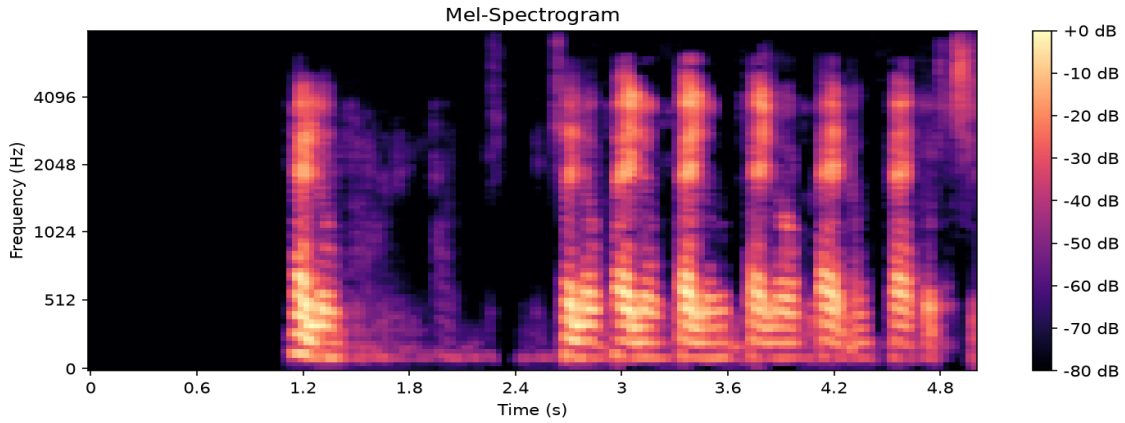
Model	Label	Confidence
CNN	spoof	0.9993817806243896
SVM	spoof	0.9921225766357823

Model	Label	Confidence
AASIST	bonafide	0.7787631899118423

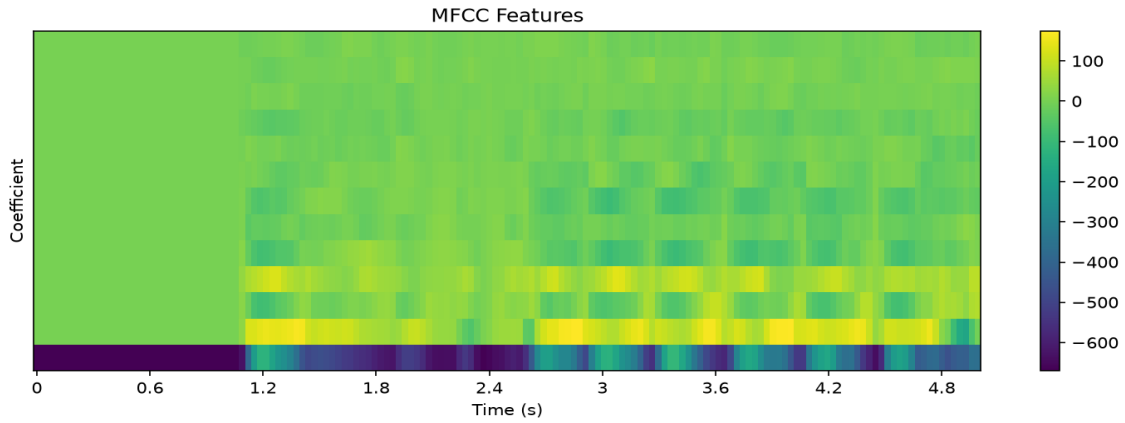
7. Visual Analysis



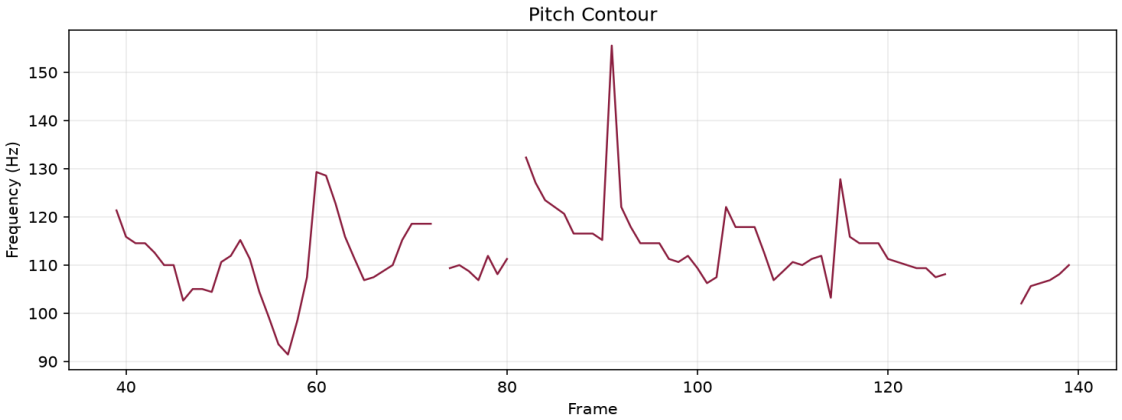
Waveform: Time-domain waveform.



Spectrogram: Frequency content over time.



MFCC: Mel-frequency cepstral coefficients.



Pitch contour: Estimated fundamental frequency.

8. Conclusion

The combined evidence indicates an overall assessment of **inconclusive** with a reported risk level of **medium**. These are indicators from the available reports and should be interpreted with the original audio and chain of custody.

9. Limitations

- Training data: All models were trained exclusively on ASVspoof 2019 LA, which contains English-language speech produced by 19 text-to-speech and voice-conversion systems. Generalisation to other languages, accents, or synthesis methods outside this distribution is not guaranteed.
- Music and non-speech audio: The pipeline is designed for speech. Music, environmental sounds, or mixed audio may produce unreliable scores and should not be submitted.
- Short recordings: The AASIST model requires a 4-second window. Files shorter than 4 seconds are zero-padded, which may reduce detection accuracy.
- Metadata: File metadata can be trivially edited or removed. Metadata findings are supporting evidence only and cannot independently confirm or deny authenticity.
- Signal features: Spectral and temporal features are influenced by recording conditions, microphone characteristics, room acoustics, and audio preprocessing. A benign artefact may appear anomalous.
- Probabilistic output: All model scores are probabilities, not binary facts. The EER-calibrated threshold for AASIST (≈81.9%) was optimised on the dev set; performance on out-of-distribution material may differ.
- Legal standing: This report is a technical aid. It does not constitute legal proof and must be reviewed by a qualified forensic expert before use in any legal or regulatory proceeding.

10. Methodology

The AudioShield pipeline combines three independent AI classifiers with metadata and signal analysis, then fuses the evidence using a rule-based risk-scoring engine.

Component	Technique	Input	Output
CNN	Convolutional neural network on log mel-spectrogram (128 mel bins, 5-second window). Architecture: FinalCNN; optimised with BCE loss and balanced-accuracy threshold selection.	5-second audio → 128×T mel-spectrogram	P(spoof); threshold ≈ 0.407

Component	Technique	Input	Output
SVM	Support Vector Machine with RBF kernel on 60-dimensional MFCC + delta + delta-delta feature vector, L2-normalised. Probability calibration via Platt scaling.	5-second audio → 60-dim MFCC features	P(spoof); threshold = 0.5
AASIST	Graph Attention Network operating directly on raw waveform (64,000 samples = 4 seconds). Retrained with WeightedRandomSampler + Focal Loss (α=0.25, γ=2.0) + CosineAnnealingLR to correct class imbalance bias.	4-second raw waveform (64k samples)	P(spoof); EER threshold ≈ 0.819
Metadata	FFprobe (container/codec) + EXIFTool (embedded tags). Checks for missing device/encoder info, timestamp inconsistencies, and unusual encoding parameters.	Audio file metadata	Risk score: low / medium / high
Signal	Spectral features: MFCCs, spectral centroid, bandwidth, rolloff, zero-crossing rate. Temporal features: RMS energy, dynamic range, silence ratio. Anomaly detection against population statistics.	Raw waveform → feature vectors	Risk score + anomaly list
Fusion	Rule-based combiner. AI risk drives the overall risk level; metadata and signal provide corroborating or mitigating evidence. A model with recall < 70 % is excluded from the primary recommendation.	AI + metadata + signal risk scores	Overall risk + recommendation

Threshold selection: all thresholds were determined on the development (dev) set only. The official evaluation set was used exclusively for final metric reporting and was never seen during threshold tuning, ensuring honest out-of-sample performance estimates.

11. Appendix

Raw AI evaluation results


```
{
  "CNN": {
    "model": "CNN",
    "checkpoint": "C:\\Users\\allen\\AudioShield-Forensics\\models\\cnn\\cnn_final_best.pth",
    "dataset": "ASVspoof 2019 LA",
    "split": "eval",
    "total_samples": 71237,
    "bonafide_samples": 7355,
    "spoof_samples": 63882,
    "best_epoch": 4,
    "best_validation_loss": 0.003000220499216719,
    "threshold": {
      "value": 0.40700000000000003,
      "source": "frozen_dev_threshold",
      "selection_method": "balanced_accuracy_with_spoof_recall_constraint",
      "selection_source": "dev_only",
      "path": "C:\\Users\\allen\\AudioShield-Forensics\\results\\cnn_final_threshold.json"
    },
    "normalization": {
      "method": "min-max",
      "train_min": -80.0,
      "train_max": 0.0,
      "source": "C:\\Users\\allen\\AudioShield-Forensics\\results\\cnn_final_norm.json"
    },
    "threshold_independent_metrics": {
      "roc_auc": 0.9800478133427984,
      "eer": 0.0716185237520802,
      "eer_percent": 7.161852375208019
    },
    "metrics": {
      "accuracy": 0.6982888105900024,
```

```
      "precision": 0.9998820726903936,
      "recall": 0.6636298174759713,
      "f1_score": 0.7977719441856964,
      "specificity": 0.9993201903467029,
      "fpr": 0.0006798096532970768,
      "tpr": 0.6636298174759713,
      "balanced_accuracy": 0.8314750039113371,
      "threshold": 0.40700000000000003,
      "selection_method": "balanced_accuracy_with_spoof_recall_constraint"
    },
    "confusion_matrix": {
      "labels": [
        "bonafide",
        "spoof"
      ],
      "matrix": [
        [
          7350,
          5
        ],
        [
          21488,
          42394
        ]
      ]
    }
  },
  "SVM": {
    "model": "SVM",
    "split": "official_eval",
    "total_samples": 71237,
    "bonafide_samples": 7355,
    "spoof_samples": 63882,
    "metrics": {
      "accuracy": 0.8940017125931748,
      "precision": 0.9543628708319218,
      "recall": 0.926082464544003,
      "f1_score": 0.9400100102485879,
      "roc_auc": 0.874909277304299,
      "eer": 0.16983042600362058
    }
  },
  "AASIST": {
    "total_samples": 71237,
    "bonafide_samples": 7355,
    "spoof_samples": 63882,
    "tn": 7242,
    "fp": 113,
    "fn": 9540,
    "tp": 54342,
    "accuracy": 0.8644945744486713,
    "precision": 0.9979248921127537,
    "recall": 0.850662158354466,
    "f1": 0.9184278796995022,
    "specificity": 0.9846363018354861,
    "fpr": 0.015363698164513936,
    "tpr": 0.850662158354466,
    "roc_auc": 0.9769212178274564,
    "eer": 0.07204162071337723,
    "threshold": 0.8189939028042915
  }
}
```


Raw metadata report


```
{
  "metadata": {
    "file_name": "b8387fa1-1ecb-4a57-baec-891c1051192d.m4a",
    "file_size": "31875",
    "format": "mov,mp4,m4a,3gp,3g2,mj2",
    "duration": "6.528063",
    "codec": "aac",
    "sample_rate": "16000",
    "channels": 1,
    "bitrate": "39062",
    "encoder": null,
    "creation_time": "2026-09-23T17:01:28.000000Z",
    "modification_time": "2026:09:23 17:01:28",
    "device_info": null,
    "software": null,
    "comments": null,
    "raw_ffprobe": {
```

```
  "streams": [
    {
      "index": 0,
      "codec_name": "aac",
      "codec_long_name": "AAC (Advanced Audio Coding)",
      "profile": "LC",
      "codec_type": "audio",
      "codec_tag_string": "mp4a",
      "codec_tag": "0x6134706d",
      "mime_codec_string": "mp4a.40.2",
      "sample_fmt": "fltp",
      "sample_rate": "16000",
      "channels": 1,
      "channel_layout": "mono",
      "bits_per_sample": 0,
      "initial_padding": 0,
      "id": "0x2",
      "r_frame_rate": "0/0",
      "avg_frame_rate": "0/0",
      "time_base": "1/16000",
      "start_pts": 0,
      "start_time": "0.000000",
      "duration_ts": 104449,
      "duration": "6.528062",
      "bit_rate": "32140",
      "nb_frames": "102",
      "extradata_size": 2,
      "disposition": {
        "default": 1,
        "dub": 0,
        "original": 0,
        "comment": 0,
        "lyrics": 0,
        "karaoke": 0,
        "forced": 0,
        "hearing_impaired": 0,
        "visual_impaired": 0,
        "clean_effects": 0,
        "attached_pic": 0,
        "timed_thumbnails": 0,
        "non_diegetic": 0,
        "captions": 0,
        "descriptions": 0,
        "metadata": 0,
        "dependent": 0,
        "still_image": 0,
        "multilayer": 0
      },
      "tags": {
        "creation_time": "2026-09-23T17:01:28.000000Z",
        "language": "und",
        "handler_name": "SoundHandler"
      }
    }
  ],
  "format": {
    "filename": "C:\\Users\\allen\\AudioShield-Forensics\\uploads\\b8387fa1-1ecb-4a57-baec-891c1051192d.m4a",
    "nb_streams": 1,
    "nb_programs": 0,
    "nb_stream_groups": 0,
    "format_name": "mov,mp4,m4a,3gp,3g2,mj2",
    "format_long_name": "QuickTime / MOV",
    "start_time": "0.000000",
    "duration": "6.528063",
    "size": "31875",
    "bit_rate": "39062",
    "probe_score": 100,
    "tags": {
      "major_brand": "mp42",
      "minor_version": "0",
      "compatible_brands": "mp41isom",
      "creation_time": "2026-09-23T17:01:28.000000Z"
    }
  }
},
"raw_exiftool": {
  "SourceFile": "C:/Users/allen/AudioShield-Forensics/uploads/b8387fa1-1ecb-4a57-baec-891c1051192d.m4a",
  "ExifToolVersion": 13.59,
  "FileName": "b8387fa1-1ecb-4a57-baec-891c1051192d.m4a",
  "Directory": "C:/Users/allen/AudioShield-Forensics/uploads",
  "FileSize": "32 kB",
  "FileModifyDate": "2026:09:23 22:56:12+05:30",
```

```
    "FileAccessDate": "2026:09:23 22:56:12+05:30",
    "FileCreateDate": "2026:09:23 22:56:12+05:30",
    "FilePermissions": "-rw-rw-rw-",
    "FileType": "M4A",
    "FileTypeExtension": "m4a",
    "MIMEType": "audio/mp4",
    "MajorBrand": "MP4 v2 [ISO 14496-14]",
    "MinorVersion": "0.0.0",
    "CompatibleBrands": [
      "mp41",
      "isom"
    ],
    "MediaDataSize": 29952,
    "MediaDataOffset": 92,
    "MovieHeaderVersion": 0,
    "CreateDate": "2026:09:23 17:01:28",
    "ModifyDate": "2026:09:23 17:01:28",
    "TimeScale": 16000,
    "Duration": "6.53 s",
    "PreferredRate": 1,
    "PreferredVolume": "100.00%",
    "PreviewTime": "0 s",
    "PreviewDuration": "0 s",
    "PosterTime": "0 s",
    "SelectionTime": "0 s",
    "SelectionDuration": "0 s",
    "CurrentTime": "0 s",
    "NextTrackID": 3,
    "TrackHeaderVersion": 0,
    "TrackCreateDate": "2026:09:23 17:01:28",
    "TrackModifyDate": "2026:09:23 17:01:28",
    "TrackID": 2,
    "TrackDuration": "6.53 s",
    "TrackLayer": 0,
    "TrackVolume": "100.00%",
    "MatrixStructure": "1 0 0 0 1 0 0 0 1",
    "MediaHeaderVersion": 0,
    "MediaCreateDate": "2026:09:23 17:01:28",
    "MediaModifyDate": "2026:09:23 17:01:28",
    "MediaTimeScale": 16000,
    "MediaDuration": "6.53 s",
    "MediaLanguageCode": "und",
    "HandlerDescription": "SoundHandler",
    "Balance": 0,
    "AudioFormat": "mp4a",
    "AudioChannels": 1,
    "AudioBitsPerSample": 16,
    "AudioSampleRate": 16000,
    "HandlerType": "Metadata",
    "EncodingTime": "2026:09:23 22:31:29+05:30",
    "AvgBitrate": "36.7 kbps"
  }
},
"summary": {
  "file_name": "b8387fa1-1ecb-4a57-baec-891c1051192d.m4a",
  "file_size": "31875",
  "format": "mov,mp4,m4a,3gp,3g2,mj2",
  "duration": "6.528063",
  "codec": "aac",
  "sample_rate": "16000",
  "channels": 1,
  "bitrate": "39062",
  "encoder": null,
  "creation_time": "2026-09-23T17:01:28.000000Z",
  "modification_time": "2026:09:23 17:01:28",
  "device_info": null,
  "software": null,
  "comments": null
},
"analysis": {
  "findings": [
    {
      "field": "device",
      "has_issue": true,
      "severity": "low",
      "description": "Device information is missing; provenance cannot be verified from metadata alone."
    },
    {
      "field": "encoder",
      "has_issue": true,
      "severity": "low",
      "description": "Encoder or software information is missing."
```

```
      }
    ],
    "risk_score": "low",
    "suspicious_fields": [],
    "missing_fields": [
      "device_info",
      "encoder",
      "software",
      "comments"
    ]
  }
}
```


Raw signal report


```
{
  "spectral_features": {
    "spectral_centroid": {
      "mean": 1772.1890855057638,
      "std": 1267.8856824414622,
      "min": 0.0,
      "max": 5074.327179941491
    },
    "spectral_bandwidth": {
      "mean": 1428.3116588401356,
      "std": 607.3785144575628,
      "min": 0.0,
      "max": 2851.386462303061
    },
    "spectral_rolloff": {
      "mean": 3277.5676751592355,
      "std": 2137.212835707652,
      "min": 0.0,
      "max": 7187.5
    },
    "spectral_flatness": {
      "mean": 0.18149975624455808,
      "std": 0.34644052489505295,
      "min": 9.069852239917964e-06,
      "max": 1.0000007152557373
    },
    "spectral_contrast": {
      "mean": 22.55968539092094,
      "std": 5.535563888421251,
      "min": 10.737564707492076,
      "max": 32.5285653348579
    }
  },
  "temporal_features": {
    "zero_crossing_rate": {
      "mean": 0.15158489251592358,
      "std": 0.13988528060912456,
      "min": 0.0,
      "max": 0.58056640625
    },
    "rms_energy": {
      "mean": 0.024278683129481643,
      "std": 0.03118444170853396,
      "min": 0.0,
      "max": 0.11663836240768433
    },
    "pitch": {
      "mean": 112.69245600283439,
      "std": 8.412881053217031,
      "min": 91.43616857570666,
      "max": 155.56349186104043
    }
  },
  "anomalies": {
    "anomalies": [],
    "risk_score": "low",
    "summary": "No broad signal-level anomalies identified."
  },
  "temporal_analysis": {
    "consistency_score": 1.0,
    "anomalies": [],
    "summary": "Temporal patterns are broadly consistent.",
    "pitch_analysis": {
      "stability_score": 1.0,
      "anomalies": [],
      "description": "Pitch contour is broadly continuous."
    },
```

```
    "energy_analysis": {
      "energy_mean": 0.024278681725263596,
      "energy_std": 0.031184442341327667,
      "anomalies": []
    }
  },
  "plots": [
    "results\\signal_plots\\waveform.png",
    "results\\signal_plots\\spectrogram.png",
    "results\\signal_plots\\mfcc.png",
    "results\\signal_plots\\pitch_contour.png"
  ],
  "signal_risk": "low",
  "summary": "Signal risk is low. No broad signal-level anomalies identified."
}
```


Methodology note

See Section 10 (Methodology) for a full description of each pipeline component, the feature extraction approach, and the threshold selection procedure. No new model inference is performed during report generation. Evidence sources remain separated to preserve interpretability.